

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,  
LONERE**

**Winter End Semester Examination – Dec. 2019**

**Course: B. Tech in Chemical Engg**

**Sem: V**

**Subject: Chemical Technology (BTCHC505)**

**Marks: 60**

**Date:-18/12/2019**

**Duration:- 3 Hr.**

**Instructions to the Students:** 1. Illustrate your answers with neat sketches, diagram etc., wherever necessary. 2. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly. 3. Attempt Any FIVE of the following

|                                                                                                                                                       | (Level/CO ) | Marks       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|
| <b>Q. 1 Solve any two of the following</b>                                                                                                            |             | <b>12 M</b> |
| (A) Explain with neat schematic representation of the electrolysis of aqueous salt solution by the mercury, diaphragm and by the membrane process     | CO1         |             |
| (B) Explain difference between Unit Operation and Unit Process with examples                                                                          |             |             |
| (C) Describe the manufacturing process of soda ash by modified Solvay process and discuss the corrosion problems encountered in this process.         |             |             |
| <b>Q.2 Solve the following</b>                                                                                                                        |             | <b>12 M</b> |
| (A) Describe the manufacturing process of following with neat labeled flow sheet                                                                      | CO2         |             |
| i) Phosphoric acid by electric furnace method                                                                                                         |             |             |
| ii) Ammonia by Hyber process                                                                                                                          |             |             |
| <b>Q. 3 Solve any two of the following</b>                                                                                                            |             | <b>12 M</b> |
| (A) Describe production of sulfur by Frasch process with a) Raw Materials b) process Description c) process flow diagram d) Major Engineering Problem | CO2         |             |
| (B) Explain the continuous hydrolysis and saponification process of soap production with respect to chemical reactions, flow diagrams                 |             |             |
| (C) With the help of neat flow sheet describe the manufacturing of sulphuric acid by contact process and discuss the uses of sulphuric acid.          |             |             |
| <b>Q. 4 Solve any two of the following</b>                                                                                                            |             | <b>12 M</b> |
| (A) Explain Extraction of sugar cane to produce crystalline white sugar with neat flow sheet ,process description, major engineering problem          | CO3         |             |
| (B) Production of Ethyl alcohol by fermentation with a) Raw Materials b) process Description c) process flow diagram d) Major Engineering Problem     |             |             |
| (C) Short note on Fermentation products from petroleum.                                                                                               |             |             |
| <b>Q. 5 Solve the following.</b>                                                                                                                      |             | <b>12 M</b> |
| (A) How will you produce pulp by Kraft pulp process w.r.t. chemical reaction, process flow diagram and major engineering problems                     | CO3         |             |

- (B) Explain different manufacturing processes for Polymer, Ethenic polymer processes, Polycondensation Processes, Polyurethanes.

12 M

**Q. 6 Solve the following.**

CO4

- (A) Describe production of crude petroleum and Refinery products.
- (B) Short note on Pyrolysis , cracking and Reforming

---

**Paper End**

---